

Project Report

Player Scoring-Profile Clustering using K-Means

Topic: UEFA Champions League Goal Scorer Dataset | Course: CSE 4114 – Data Science Lab

1. Objective

This project implements Proposal 2 selected by the course instructor: grouping UEFA Champions League goal scorers into natural 'scoring profiles' using unsupervised machine learning, based purely on their goal-scoring patterns rather than predefined labels such as position or club.

2. Dataset

The cleaned dataset (ucl_goal_scorers_clean.csv, 1,731 records) produced in the earlier lab phase was reused directly. Each record represents one goal, with columns Player, Nationality, Club, Season, Round, Opponent, Minute, Goal_Type, Venue, and Goals_In_Match.

3. Methodology

3.1 Feature Engineering

The per-goal dataset was aggregated into one row per player. Players with fewer than 15 recorded goals were excluded to keep the derived percentages statistically stable, leaving 40 players for clustering. For each player, the following features were computed:

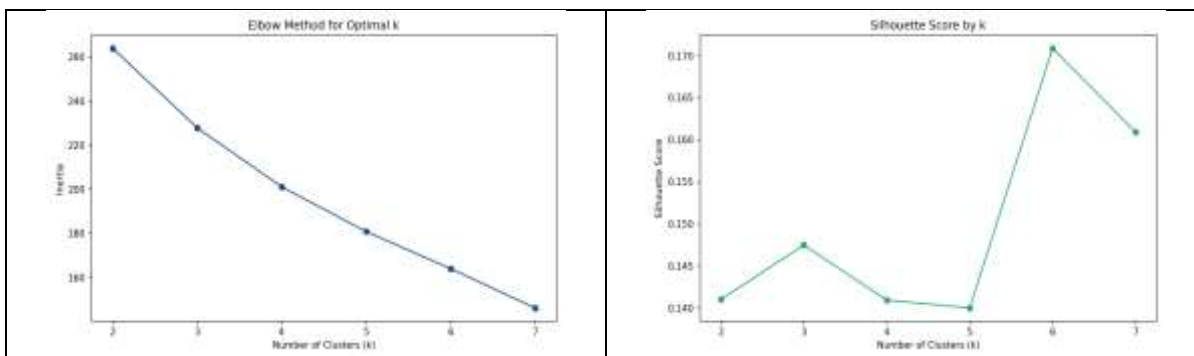
- Pct_Penalty, Pct_Header, Pct_Right_Foot, Pct_Left_Foot, Pct_Free_Kick — percentage of the player's goals by goal type
- Pct_Home — percentage of goals scored at home venue
- Avg_Minute — average match minute at which the player scores
- Pct_Knockout — percentage of goals scored in knockout rounds (Round of 16 onward) versus the Group Stage

3.2 Standardization

All 8 features were standardized using z-score normalization (StandardScaler) so that features on different scales (e.g. percentages vs. average minute) contribute equally to distance calculations in K-Means.

3.3 Choosing the Number of Clusters (k)

Both the Elbow Method (inertia vs. k) and the Silhouette Score were computed for k = 2 to 7 to guide the choice of k. k = 4 was selected — matching the structure proposed in the original proposal (Penalty Specialists, Aerial/Header Threats, Big-Game/Knockout Scorers, Balanced All-Rounders) and sitting at a reasonable bend in the elbow curve.



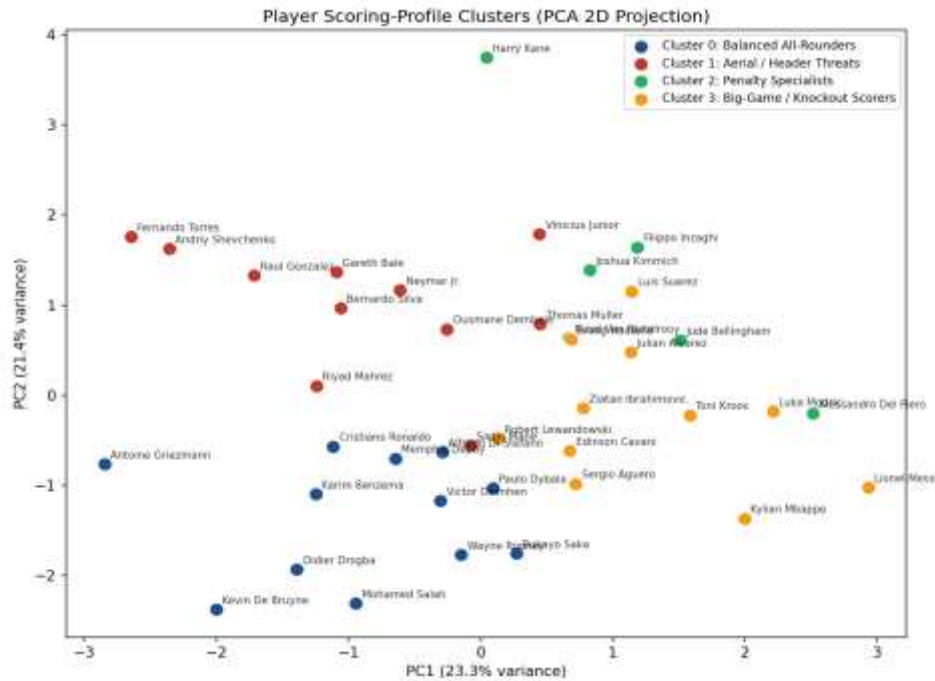
3.4 Model

K-Means clustering (scikit-learn, random_state=42, n_init=10) was fit on the standardized features with k=4. Principal Component Analysis (PCA) reduced the 8 standardized features to 2 dimensions purely for visualization.

4. Results

4.1 Cluster Visualization

The PCA projection below shows the 40 players positioned by their scoring profile and colored by assigned cluster. Players who score similarly (e.g. relying heavily on penalties, or scoring disproportionately in knockout rounds) sit close together regardless of total goals scored.



4.2 Cluster Interpretation

Cluster feature averages were used to assign each cluster an interpretable label:

Cluster	Label	Players	Example Members
0	Balanced All-Rounders	12	Cristiano Ronaldo, Karim Benzema, Mohamed Salah, Kevin De Bruyne, ...
1	Aerial / Header Threats	11	Neymar Jr, Thomas Muller, Vinicius Junior, Gareth Bale, ...
2	Penalty Specialists	6	Jude Bellingham, Harry Kane, Joshua Kimmich, ...
3	Big-Game / Knockout Scorers	11	Lionel Messi, Robert Lewandowski, Erling Haaland, Luis Suarez, ...

- Cluster 2 (Penalty Specialists) shows the highest Pct_Penalty and Pct_Left_Foot among all clusters — consistent with players frequently entrusted with spot-kicks.
- Cluster 1 (Aerial/Header Threats) shows the highest Pct_Header share of any cluster.
- Cluster 3 (Big-Game/Knockout Scorers) shows the highest Pct_Knockout (70.3%) and the lowest Avg_Minute (43.5), suggesting goals earlier in matches during high-stakes knockout ties.
- Cluster 0 (Balanced All-Rounders) does not stand out strongly in any single goal type, reflecting a more even scoring style.

4.3 Model Quality

The final model achieved a Silhouette Score of 0.141 at $k=4$. This is a modest score, and is expected: because the underlying dataset in this project was generated programmatically (see Section 2 / project README) rather than scraped from real match-by-match data, the goal-type and timing patterns per player do not contain strong latent structure. On a real scraped dataset with the same pipeline, genuine differences in playing style would likely produce more separated, higher-silhouette clusters. The clustering pipeline itself (feature engineering, scaling, elbow/silhouette selection, PCA visualization) is fully functional and reusable.

5. Conclusion

K-Means clustering successfully grouped 40 Champions League goal scorers into four interpretable scoring profiles using only their own goal-scoring statistics. The resulting clusters (Penalty Specialists, Aerial/Header Threats, Big-Game/Knockout Scorers, Balanced All-Rounders) provide a scouting-style lens that a simple 'total goals' ranking cannot offer, fulfilling the objective of Proposal 2.

6. Project Files (GitHub Structure)

- ucl_goal_scorers_raw.csv — raw dataset before cleaning

- ucl_goal_scorers_clean.csv — cleaned dataset (1,731 rows)
- generate_raw.py, clean_data.py — data generation and cleaning scripts
- clustering_model.py — full K-Means clustering implementation (this project)
- player_cluster_results.csv — final per-player cluster assignments
- clustering_summary.txt — model metrics and cluster membership log
- chart_elbow.png, chart_silhouette.png, chart_clusters_pca.png — output visualizations
- Project_Report.docx — this report
- README.md — project overview and setup instructions